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DSA1080 XA PROJECT REPORT

INTRODUCTION

This report presents an end-to-end analysis of a European retail bank's customer base, covering data cleaning, exploratory analysis, statistical summarization, visualization, and a supervised machine learning approach to churn prediction.

The goal is to translate raw customer records into concrete, actionable findings that a retention team could act on immediately. The analysis was carried out entirely in Python, using pandas and NumPy for data manipulation, seaborn and matplotlib for visualization, and scikit-learn for predictive modeling. The full analysis script, along with all supporting charts, accompanies this report.

PROBLEM STATEMENT

The bank is losing customers, but it doesn't know exactly who is leaving or the main reasons why. Because of this blind spot, the bank spends money trying to keep everyone. This wastes resources on loyal customers who were never going to leave, while the people actually at risk of walking away are ignored.

This project aims to fix that by answering three key questions:

- **Who is leaving the most?** We will look at where customers live, their personal details, the bank products they use, and how active they are to identify which specific groups are quitting the bank.
- **Can a computer predict who will leave next?** We want to test if an AI program (machine learning) can look at a customer's basic account data and accurately guess if they are about to close their account.
- **What should we do to stop them?** Based on the patterns the data reveals, what specific actions should the bank take to convince these at-risk customers to stay?

By answering these questions, the bank can stop scrambling *after* customers decide to leave and instead step in early with targeted offers to keep them.

DATA DESCRIPTION

The dataset used is the publicly available Bank Customer Churn Modelling dataset, sourced from Kaggle and originally derived from European retail banking records. It contains 10,000 customer records across three markets: France, Germany, and Spain.

Field	**Description**
Credit Score	Customer's credit score (300–850 range)
Geography	Customer's country: France, Germany, or Spain
Gender	Customer's gender
Age	Customer's age in years
Tenure	Number of years as a bank customer
Balance	Current account balance
NumOfProducts	Number of bank products held (1–4)
HasCrCard	Whether the customer holds a credit card (1/0)
IsActiveMember	Whether the customer is an active member (1/0)
EstimatedSalary	Customer's estimated annual salary
Exited	Target variable: 1 = churned, 0 = retained

METHODOLOGY

- **Removed unnecessary labels:** We dropped columns like row numbers, customer IDs, and last names because they are just identification numbers and names—they don't help us figure out if someone will leave.
- **Checked for duplicates:** We made sure no customer was listed twice. None were, but we set up a rule to automatically remove duplicates if any pop up later.
- **Checked for missing information:** All fields were filled out. However, we added a safety rule that automatically fills in any missing future data using average or common values.

- **Cleaned up text formatting:** We fixed formatting errors in text fields like location (*Geography*) and *Gender* by removing extra spaces, ensuring the computer reads them consistently.

Feature Engineering (Creating New Helpful Signals)

- **Savings relative to income (*Balance-to-Salary Ratio*):** We divided a customer's balance by their salary to see how much of their overall wealth is sitting in the bank. (We added 1 to the salary so the system won't crash if someone's income is reported as zero.)
- **Age categories (*Age Group*):** Instead of looking at exact ages, we grouped customers into clear life stages—Young Adult, Middle Aged, Senior, and Elderly—to easily spot patterns across different age groups.

EXPLORATORY DATA ANALYSIS

Descriptive statistics (mean, median, standard deviation) were computed for all numeric fields. Churn rate was then broken down by Geography, Gender, Number of Products, and Active Membership status using group-by aggregation, in order to surface which segments carry disproportionate churn risk.

MACHINE LEARNING APPROACH

A Decision Tree Classifier (max depth 5) was trained to predict the Exited target variable. Categorical variables were encoded (one-hot encoding for the nominal Geography field, binary encoding for Gender) and numeric features were standardized using StandardScaler.

Data was split 80/20 into training and test sets using stratified sampling (stratify=model_target) to preserve the 79.6%/20.4% retained/churned ratio in both sets. Model performance was evaluated using accuracy, precision, recall, and F1-score, with particular attention to recall given the class imbalance in the target variable.

RESULTS

Of the 10,000 customers analyzed, 20.4% exited the bank, while 79.6% were retained. This baseline rate is the reference point for all segment-level comparisons below.

Churn by Geography

Field	**Churn Rate**
France	16.2%
Germany	32.4%
Spain	16.7%

Germany shows a churn rate roughly double that of France and Spain, making it the single strongest geographic risk signal in the dataset.

Churn by Number of Products

Number Of Products	**Churn Rate**
1	27.7%
2	7.6%
3	82.7%
4	100%

Customers holding two products churn the least of any group, but churn rises sharply for customers holding three or four products. This counter-intuitive pattern suggests a breakdown in the multi-product customer experience rather than a benefit of deeper engagement.

Churn by Active Membership

Membership Status	**Churn Rate**
Inactive (0)	26.9%
Active (1)	14.3%

Inactive members churn at nearly twice the rate of active members, confirming engagement level as a strong and continuously observable early-warning signal.

Model Performance

Metric	Value
Accuracy	85.6%
Precision	78.7%
Recall	40.0%
F1-Score	0.531

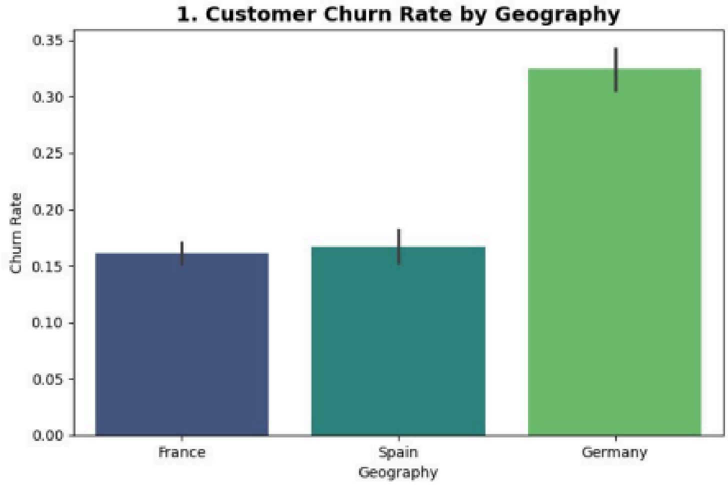
- **Overall Score (85.6% Accuracy):** The model correctly guesses whether a customer stays or leaves about 86% of the time overall.
- **When It Flags a Leaver (78.7% Precision):** When the model predicts that a customer is about to leave, it gets it right roughly 4 out of 5 times. It rarely raises a false alarm on a customer who is staying.
- **Catching the Real Leavers (40.0% Recall):** Here is the main problem: the model only catches 4 out of every 10 customers who *actually* end up leaving, missing the other 6.
- **Why This Happens:** Most customers in the data stayed with the bank (about 80%), while only a few left (about 20%). Because staying is so common, the model naturally leans toward guessing "staying" to play it safe.

Key Takeaway:

High overall accuracy can be misleading. The biggest takeaway is that future versions of the model must focus on catching more of the real leavers, rather than just keeping a high overall score.

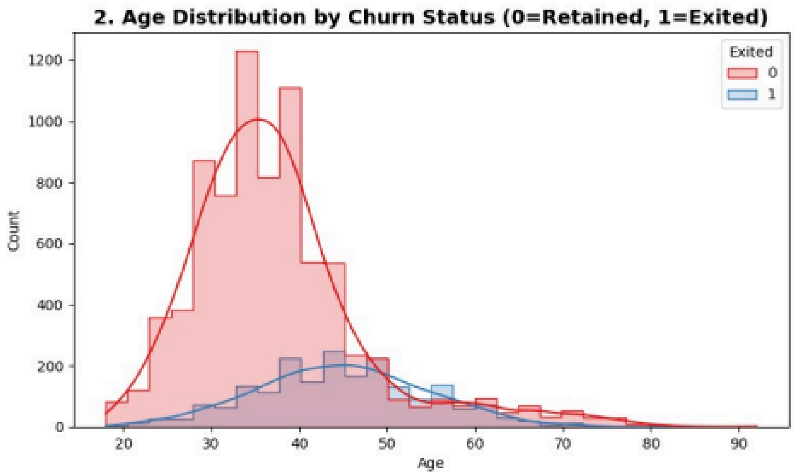
VISUALIZATION

Chart 1: Customer Churn Rate by Geography



Germany's churn rate is approximately double that of France and Spain.

Chart 2: Age Distribution by Churn Status

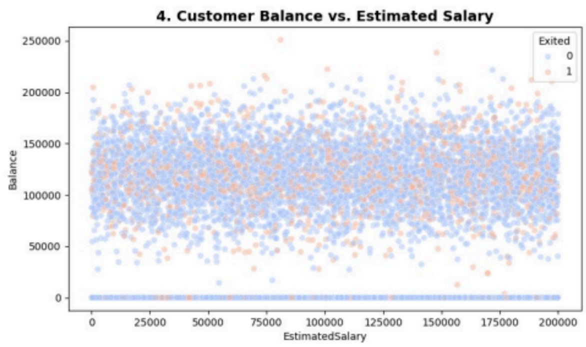


Churned customers (Exited = 1) skew noticeably older than retained customers.

Chart 3: Credit Score Distribution by Churn Status

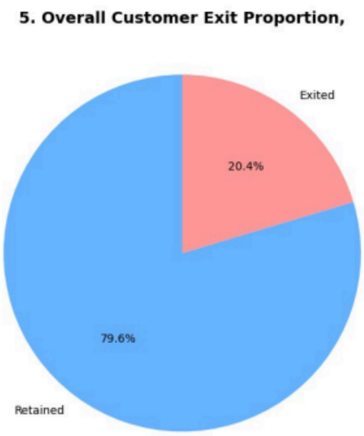
Credit score distributions are broadly similar between retained and exited customers, suggesting limited standalone predictive power.

Chart 4: Customer Balance vs. Estimated Salary



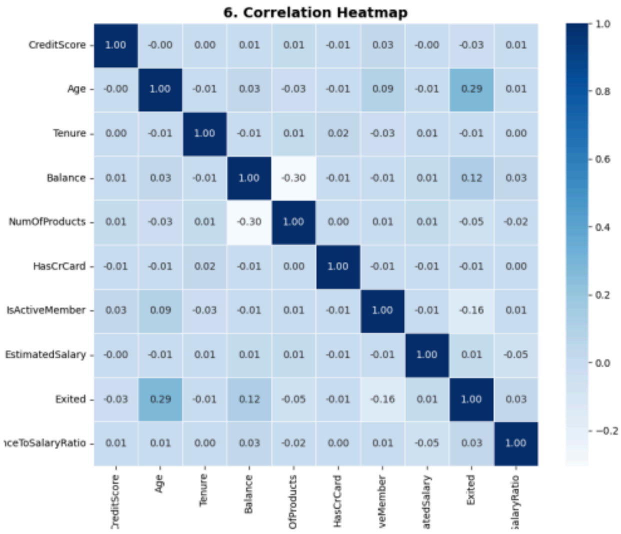
No strong linear relationship is visible between balance, salary, and churn status; churned customers (orange) are distributed across the full range of both variables.

Chart 5: Overall Customer Exit Proportion



79.6% of customers were retained; 20.4% exited — the class imbalance addressed in the modeling stage.

Chart 6: Correlation Heatmap of Numeric Variables



Age shows the strongest positive correlation with churn among numeric variables, though no single numeric feature is strongly correlated on its own — consistent with churn being driven by a combination of factors.

CONCLUSIONS

This study set out to answer three core questions: **who** is leaving the bank, **can a computer predict** their departure using basic account data, and **what steps** should the bank take next? The data gave clear answers to all three.

Key Drivers of Customer Departure

- **Location:** Customers in Germany are leaving at roughly double the rate of those in France or Spain.
- **Number of Products:** Counterintuitively, customers with 3 or 4 bank products leave at much higher rates than those with only 1 or 2. Having more products does not automatically equal greater loyalty.
- **Customer Activity:** Inactive members quit nearly twice as fast as active ones. A drop in activity is our strongest early warning signal.

Predictive Model Performance

While the computer model scored **85.6% overall accuracy** and **78.7% precision** (meaning its warnings are rarely false alarms), it only achieved **40.0% recall**.

This means the model currently misses more than half of the people who actually end up leaving. Because 8 out of 10 customers stay, the model gets lazy and defaults to guessing "staying." Going forward, the bank must judge future models by how many true leavers they catch (recall), not just overall accuracy.

Recommended Action Plan

- **Fix the German Market:** Focus retention efforts in Germany. Investigate local competitor rates, pricing, and customer service to understand why German accounts are dropping off so rapidly.
- **Overhaul Product Bundles:** Investigate what is frustrating customers who hold 3 or 4 products. This is the single biggest churn trigger found in the entire study.
- **Re-engage Inactive Accounts:** Set up automated campaigns to reach out to customers as soon as their activity drops, long before they decide to close their accounts completely.

Limitations

- **Missing Context:** The dataset lacks customer service history (e.g., complaint logs or wait times), telling us *who* is leaving, but not the exact personal frustrations *why*.
- **Imbalanced Data:** Because 80% of customers stayed and only 20% left, the model naturally learned to focus on the majority class rather than catching leavers.

Next Steps & Improvements

- **Try Smarter AI Models:** Test advanced algorithms like Random Forest or XGBoost, which handle complex patterns—like the sudden jump in churn among multi-product holders—much better than a single decision tree.
- **Set Up a Benchmark:** Train a simpler, traditional model (like Logistic Regression) to see if a basic baseline performs just as well or better.
- **Balance the Training Data:** Apply statistical balancing techniques (like SMOTE or weighted class penalties) to force the model to focus heavily on catching actual churners.

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